Ain Shams University

Faculty of Engineering
Computer and Systems Engineering Department

CSE356: Internet of Things

IoT Project Report

Smart Home Intrusion Detection System

Part 2: Prototype Implementation Using Cisco Packet Tracer

Submitted by:

Omar Tarek Mosaad	**23P0278**
Omar Montaser	**23P0171**
Mohamed Ahmed	**23P0282**
Salma Mousa	**23P0023**

Instructor: Dr. Islam Tharwat Abdel Halim

Contents

List of Figures **2**

1 Introduction **3**
1.1 Purpose of the Project . . . 3
1.2 Objective of Part 2 . . . 3

2 System Topology / Simulation Setup **4**
2.1 Network Overview . . . 4
2.2 Components Used . . . 5
2.3 Connectivity Summary . . . 6
2.4 Final Simulation Topology . . . 6

3 Device Configuration **7**
3.1 Home Gateway Configuration . . . 7
3.2 End Device Configuration . . . 8
3.2.1 Smartphone . . . 8
3.2.2 Laptop (CCTV Workstation) . . . 9
3.3 Server Configuration . . . 10
3.4 IoT Device Registration . . . 10
3.5 CCTV Monitoring Interface . . . 12

4 Automation Rules / Conditions **13**
4.1 Overview of Automation Logic . . . 13
4.2 Implemented Automation Rules . . . 14
4.3 Rule Details . . . 16
4.3.1 Intrusion Alert Rules (cond1 – cond5) . . . 16
4.3.2 Camera Activation Rules (cond6, cond7, cond14, cond15) . . . 16
4.3.3 Access Control Rule (cond8) . . . 17
4.3.4 Reset / Deactivation Rules (cond9 – cond13) . . . 17

5 Testing and Results **18**
5.1 Testing Methodology . . . 18
5.2 Test Case 1: Motion Detector 1 Trigger . . . 19
5.3 Test Case 2: Motion Detector 2 Trigger . . . 20
5.4 Test Case 3: Smart Door Unlock Trigger . . . 21
5.5 Test Case 4: Window Trigger . . . 22
5.6 Test Summary . . . 23

6 Challenges and Solutions **24**
6.1 Laptop Wireless Connectivity Issue . . . 25
6.2 IoT Device Registration Challenges . . . 26
6.3 Automation Rule Logic Verification . . . 26
6.4 Summary of Challenges and Resolutions . . . 27

7 Conclusion **28**

List of Figures

1 Final SHIDS Simulation Topology in Cisco Packet Tracer 6
2 Home Gateway Wireless Configuration and Network Settings 7
3 Smartphone Network Configuration and DHCP Assignment 8
4 Laptop Network Configuration and DHCP Assignment 9
5 IoT Monitor Login Interface for Home Gateway Access 10
6 Registered IoT Devices on Home Gateway 11
7 CCTV Monitoring Interface for Security Cameras 12
8 Complete Automation Rules Configured for SHIDS 15
9 Motion Detection Rule for Alarm Activation 16
10 Automated Door Lock Control During Alarm Activation 17
11 Motion Detector 1 Triggered: Siren and Camera 1 Activated 19
12 Motion Detector 2 Triggered: Siren and Camera 2 Activated 20
13 Smart Door Unlock Trigger: Siren and Camera 1 Activated 21
14 Window Trigger: Siren and Camera 2 Activated 22
15 Installation of WPC300N Wireless Adapter on Laptop 25

1 Introduction

The rapid advancement of Internet of Things (IoT) technologies has opened new possibilities for intelligent home security systems. By interconnecting sensors, actuators, cameras, and smart devices through a unified network, IoT enables real-time monitoring and automated response mechanisms that were previously unavailable in traditional security setups.

This report documents **Part 2** of the Smart Home Intrusion Detection System (SHIDS) project, which focuses on the prototype implementation of the system designed in Part 1. The prototype is developed using **Cisco Packet Tracer**, a network simulation environment that supports IoT device modelling, wireless connectivity, and automation rule configuration.

1.1 Purpose of the Project

The Smart Home Intrusion Detection System (SHIDS) is an IoT-based residential security solution designed to detect unauthorized entry, monitor critical zones in real time, and trigger automated responses upon intrusion. The system aims to provide homeowners with immediate alerts, remote monitoring capabilities, and automated security actions — all accessible through a connected mobile or web interface.

1.2 Objective of Part 2

The primary objective of Part 2 is to validate the design proposed in Part 1 by constructing a functional simulation of SHIDS in Cisco Packet Tracer. The specific objectives of this phase are:

- Simulate the SHIDS network topology, including all IoT devices, networking infrastructure, and cloud components, within Cisco Packet Tracer.
- Configure all IoT devices, including sensors, cameras, actuators, and smart locks, and register them with the Home Gateway.
- Define and implement automation rules that govern the system's intrusion detection and alert behaviour.
- Perform functional tests to verify that the system responds correctly to simulated intrusion events, including motion detection, door and window breaches, and smoke detection.
- Document the simulation process, configuration steps, encountered challenges, and their corresponding solutions.

Together, these objectives demonstrate the practical feasibility of the SHIDS design and confirm that the proposed IoT architecture can be realised in a simulated environment prior to physical deployment.

2 System Topology / Simulation Setup

This section describes the network topology and simulation environment constructed in Cisco Packet Tracer to prototype the Smart Home Intrusion Detection System (SHIDS). The simulation replicates the physical and logical architecture proposed in Part 1, covering all IoT sensing devices, actuators, networking infrastructure, and cloud-side components.

2.1 Network Overview

The simulated topology is organised around a central **Home Gateway (DLC100)**, which acts as the IoT controller and wireless access point for all home devices. The gateway bridges the local home network with the external network, represented by a cloud component, a managed switch, and a backend server. All IoT devices communicate wirelessly with the gateway, while the cloud infrastructure is connected via a wired backbone through the switch.

The topology is divided into two logical zones:

- **Home Zone** — Contains all IoT sensing and actuation devices connected wirelessly to the Home Gateway.
- **Cloud/Infrastructure Zone** — Contains the network switch, backend server, and cloud component connected via Ethernet to the Home Gateway.

2.2 Components Used

Table 1 lists all components included in the SHIDS simulation, along with their role in the system.

Table 1: Simulation Components and Their Roles

Component	Device Type	Role in SHIDS
Home Gateway (DLC100)	IoT Gateway	Central controller; manages all IoT devices, DHCP, Wi-Fi, and cloud communication
Motion Detector 1	IoT Sensor	Detects motion at the front entry zone
Motion Detector 2	IoT Sensor	Detects motion at the rear/secondary zone
Smart Door Lock	IoT Actuator	Controls main door access; locks automatically on intrusion
Webcam (cam1)	IoT Camera	Activated upon motion detection at zone 1
Webcam (cam2)	IoT Camera	Activated upon motion detection at zone 2
Window Sensor	IoT Sensor	Monitors window open/closed state
Smoke Detector	IoT Sensor	Detects smoke or fire conditions
Siren	IoT Actuator	Produces audible alarm upon intrusion confirmation
Smartphone	End Device	Homeowner mobile interface for monitoring and control
Laptop (CCTV)	End Device	Admin/CCTV monitoring workstation
Switch (2960-24TT)	Network Device	Connects gateway to server and cloud
Server (Server-PT)	Backend Server	Hosts HTTP and DNS services; simulates cloud backend
Cloud (Cloud-PT)	Cloud Component	Simulates external cloud connectivity and data routing

2.3 Connectivity Summary

- All IoT devices (sensors, cameras, actuators, smartphone, and laptop) connect **wirelessly** to the Home Gateway via Wi-Fi.
- The Home Gateway connects **via Ethernet** to the 2960-24TT Switch.
- The Switch connects via Ethernet to both the **Server-PT** and the **Cloud-PT**, forming the wired infrastructure backbone.
- The Server hosts **DNS** and **HTTP** services, with the domain `shids.com` resolving to the server's IP address.

2.4 Final Simulation Topology

Figure 1 presents the final simulated topology of SHIDS as constructed in Cisco Packet Tracer, showing all devices, their connections, and the overall network layout.

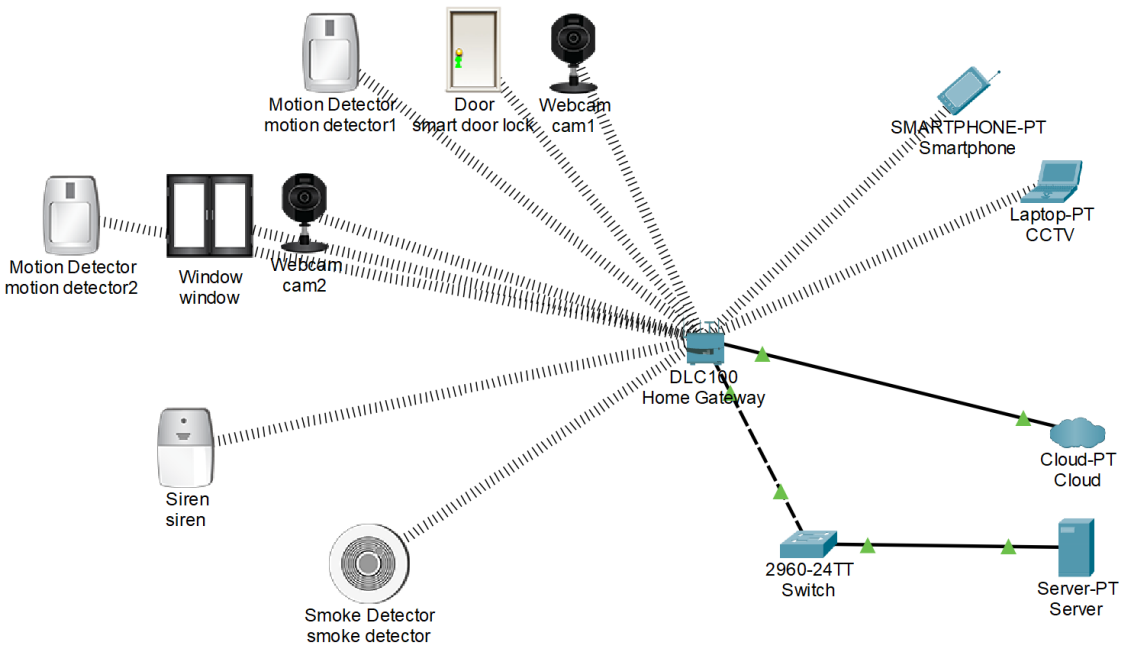


Figure 1: Final SHIDS Simulation Topology in Cisco Packet Tracer

3 Device Configuration

This section details the configuration applied to each key component of the SHIDS simulation in Cisco Packet Tracer. Proper configuration of the Home Gateway, end devices, and backend server is essential to ensure correct device registration, network connectivity, and IoT automation functionality.

3.1 Home Gateway Configuration

The **Home Gateway (DLC100)** serves as the central controller of the entire SHIDS simulation. It was configured with the following settings:

- **Wi-Fi SSID:** A dedicated wireless network name was assigned to identify the home network, allowing all IoT devices to connect wirelessly.
- **Wi-Fi Password:** A password was set to restrict access to authorised devices only.
- **DHCP:** Enabled on the gateway to automatically assign IP addresses to all connected devices, eliminating the need for manual IP configuration.
- **IoT Server / Registration:** The gateway's built-in IoT server was enabled, allowing all IoT devices to register and be managed centrally through the gateway's IoT monitor interface.

Figure 2 shows the Home Gateway wireless configuration and network settings as applied in the simulation.

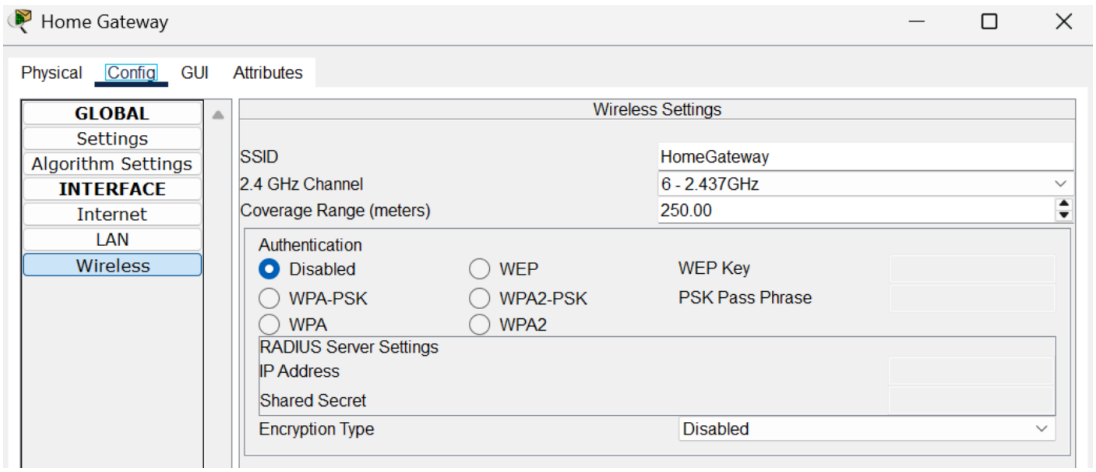


Figure 2: Home Gateway Wireless Configuration and Network Settings

3.2 End Device Configuration

3.2.1 Smartphone

The **Smartphone** represents the homeowner's mobile interface for monitoring and controlling the SHIDS system remotely. It was configured as follows:

- Connected wirelessly to the Home Gateway using the configured SSID and password.
- **DHCP** was enabled, and the gateway automatically assigned it an IP address upon connection.
- The smartphone accesses the IoT monitor interface through the gateway to view device status, receive alerts, and issue control commands.

Figure 3 shows the smartphone's network configuration and the DHCP-assigned IP address.

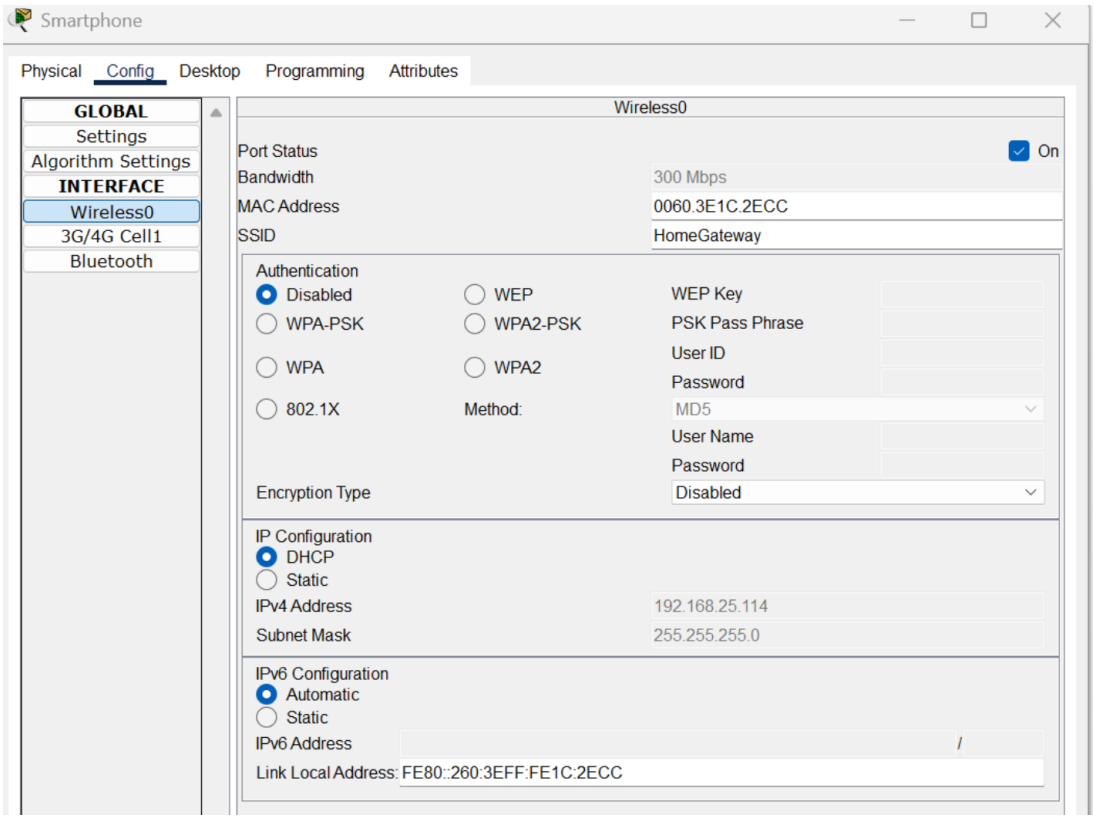


Figure 3: Smartphone Network Configuration and DHCP Assignment

3.2.2 Laptop (CCTV Workstation)

The **Laptop** serves as the CCTV monitoring and administrative workstation. It was configured as follows:

- A **WPC300N wireless network adapter** was installed on the laptop to enable Wi-Fi connectivity, as the default laptop model in Cisco Packet Tracer does not include a wireless interface.
- Connected wirelessly to the Home Gateway using the same SSID and password.
- DHCP was enabled to obtain an IP address automatically from the gateway.
- Used to access the CCTV monitoring interface and the IoT monitor dashboard via a web browser.

Figure 4 shows the laptop's wireless adapter installation and network configuration as applied in the simulation.

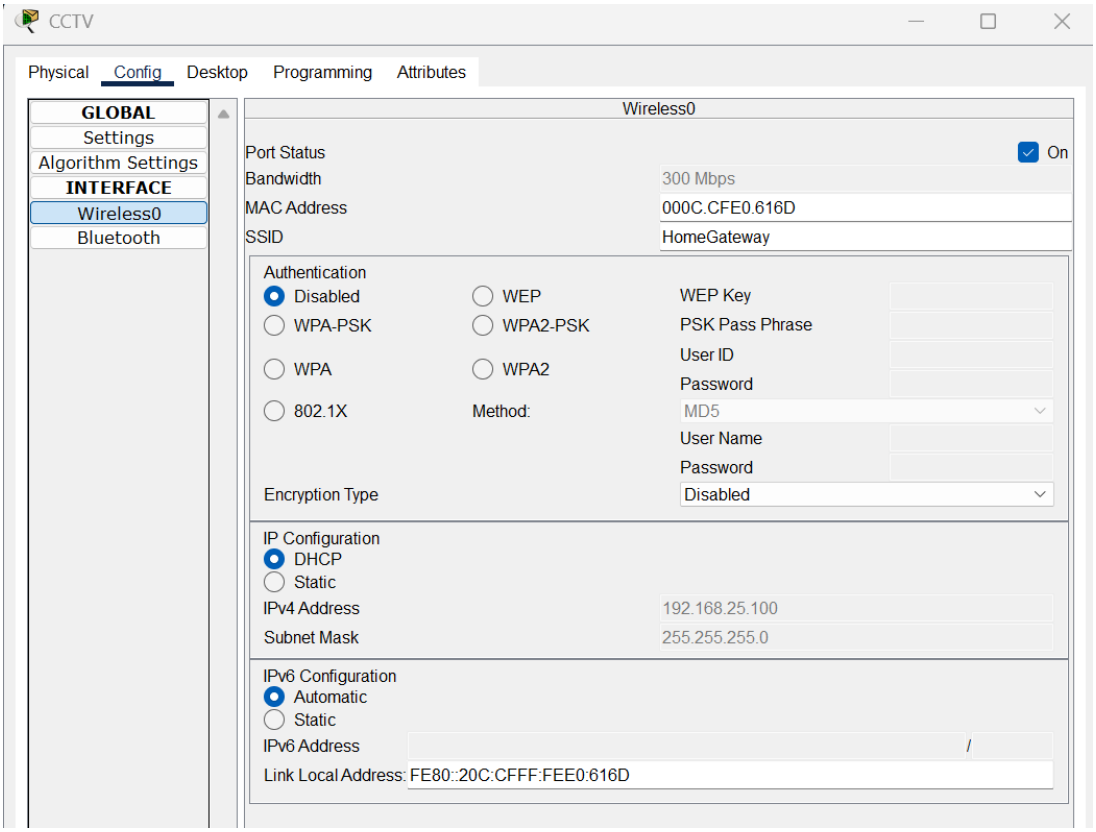


Figure 4: Laptop Network Configuration and DHCP Assignment

3.3 Server Configuration

The **Server-PT** simulates the cloud backend of SHIDS and was configured with the following services:

- **HTTP Service:** Enabled to serve web-based content and simulate the SHIDS web application interface accessible by end devices.
- **DNS Service:** Enabled and configured with the domain record `shids.com`, which resolves to the server's assigned IP address. This allows end devices to access the server using the domain name rather than a raw IP address.

3.4 IoT Device Registration

All IoT devices — including motion detectors, webcams, the smart door lock, the window sensor, the smoke detector, and the siren — were registered with the Home Gateway's built-in IoT server. Registration was performed by configuring each device's IoT server address to point to the gateway's IP address, enabling centralised management and automation rule execution.

Figure 5 shows the IoT Monitor login interface used to access the Home Gateway's device management panel.

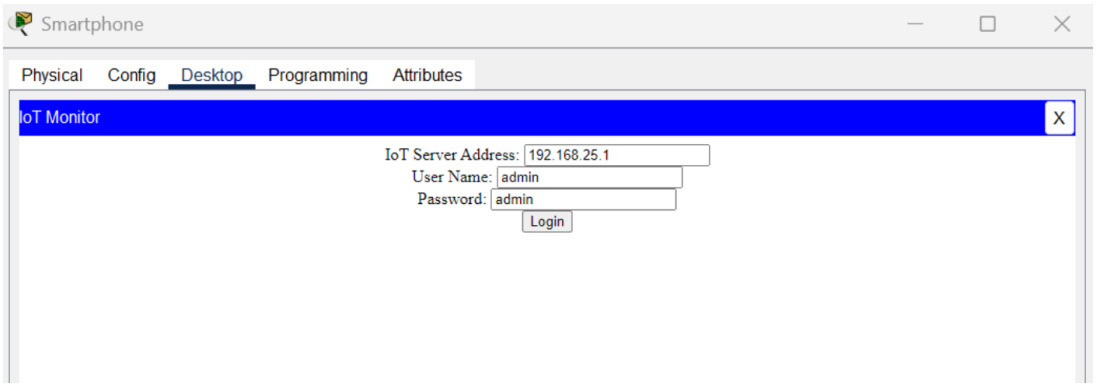


Figure 5: IoT Monitor Login Interface for Home Gateway Access

Figure 6 shows all IoT devices successfully registered and visible on the Home Gateway's IoT Monitor dashboard.

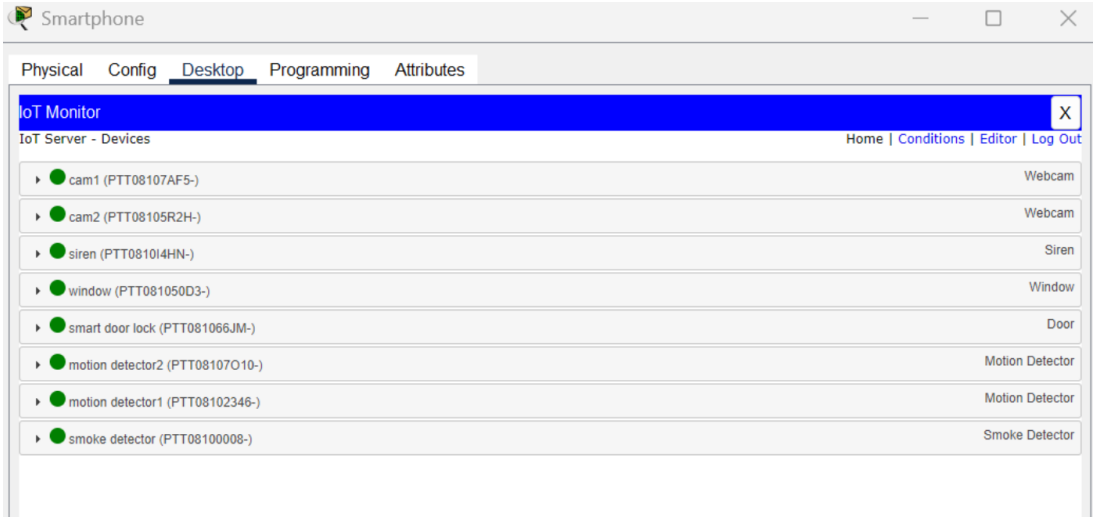


Figure 6: Registered IoT Devices on Home Gateway

3.5 CCTV Monitoring Interface

The CCTV monitoring interface is accessible from the laptop workstation and provides a live view of all connected webcams within the SHIDS simulation. This interface allows the security administrator or homeowner to visually monitor all surveilled zones in real time.

Figure 7 shows the CCTV monitoring interface as accessed from the laptop during the simulation.

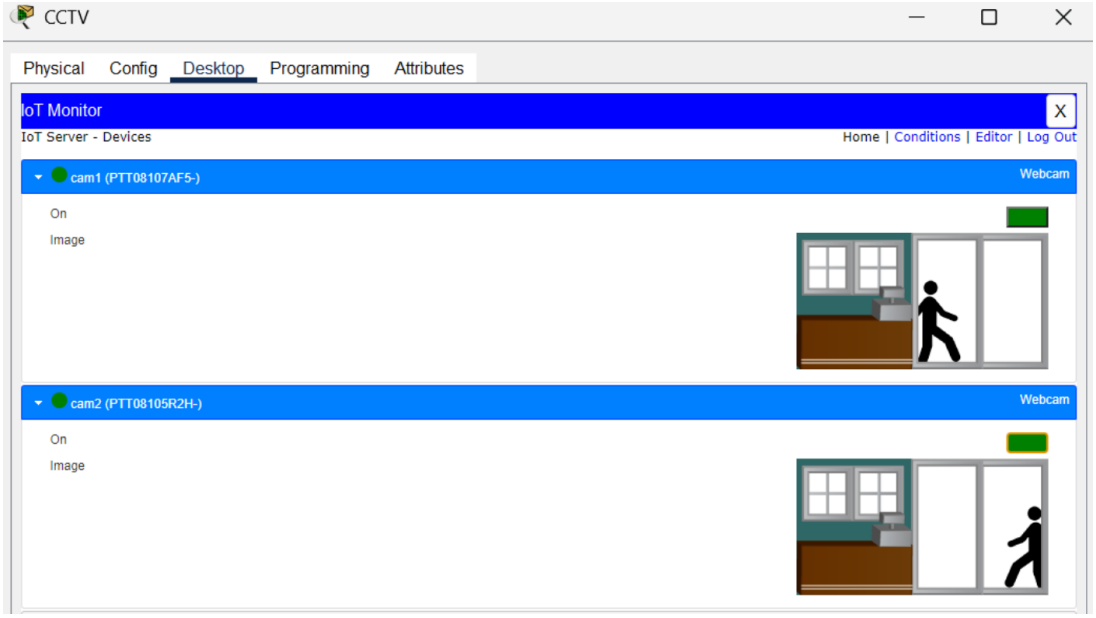


Figure 7: CCTV Monitoring Interface for Security Cameras

4 Automation Rules / Conditions

This section presents the automation rules configured within the SHIDS simulation in Cisco Packet Tracer. These rules define the conditional logic that governs the system's automated response to intrusion events, forming the core intelligence of the simulation. All rules were implemented through the Home Gateway's IoT automation interface and are executed locally at the edge layer upon sensor triggering.

4.1 Overview of Automation Logic

The automation logic of SHIDS follows a condition-action model: when a sensor detects an event, one or more actuators or devices are triggered automatically without requiring manual intervention. The rules are grouped into four categories:

- **Intrusion Alert Rules** — Activate the siren upon detection by any sensor.
- **Camera Activation Rules** — Turn on the corresponding webcam when a motion detector, door, or window is triggered.
- **Access Control Rules** — Automatically unlock the smart door lock when the siren is activated.
- **Reset / Deactivation Rules** — Turn off cameras and siren when sensors return to their normal idle state.

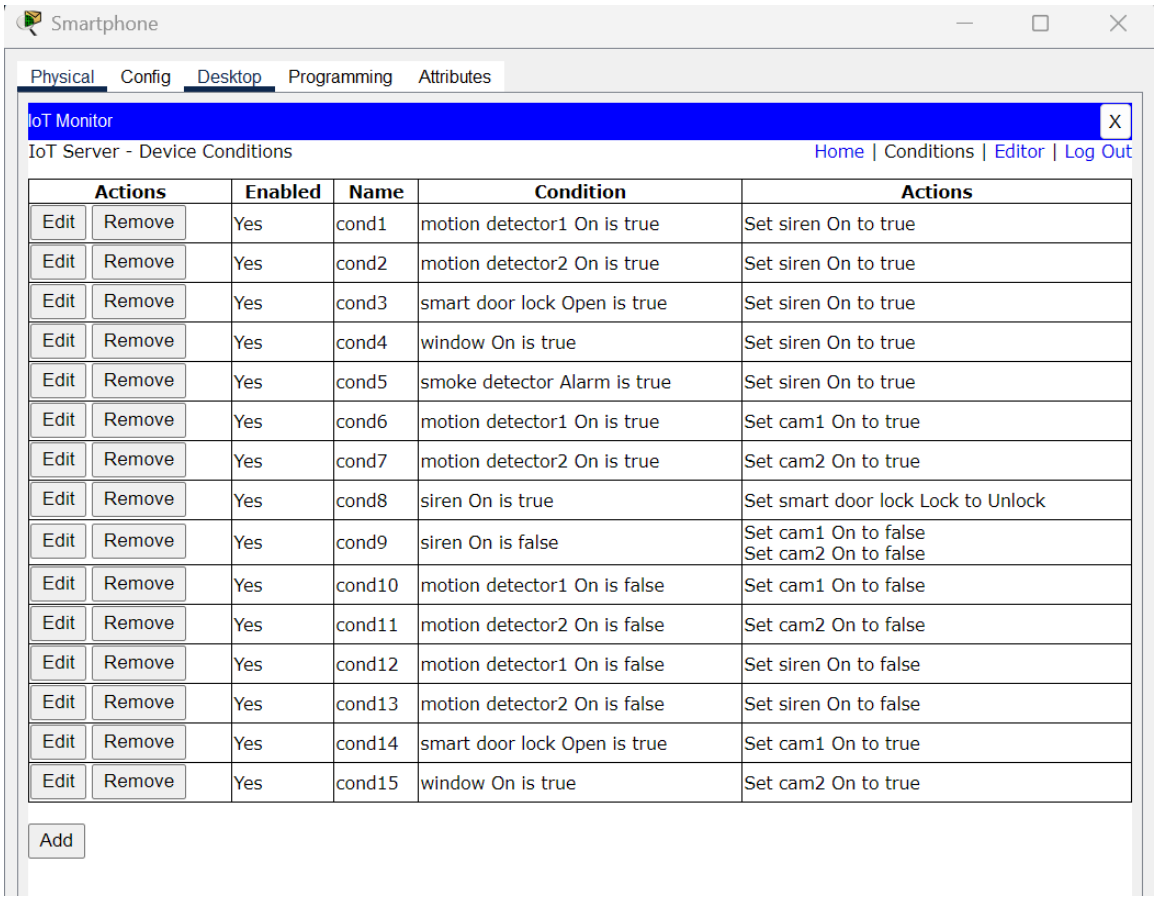
4.2 Implemented Automation Rules

Table 2 provides a complete summary of all 15 automation rules implemented in the SHIDS simulation.

Table 2: SHIDS Automation Rules

Rule	Condition (Trigger)	Action (Response)
cond1	Motion Detector 1 On is true	Set siren On to true
cond2	Motion Detector 2 On is true	Set siren On to true
cond3	Smart Door Lock Open is true	Set siren On to true
cond4	Window On is true	Set siren On to true
cond5	Smoke Detector Alarm is true	Set siren On to true
cond6	Motion Detector 1 On is true	Set cam1 On to true
cond7	Motion Detector 2 On is true	Set cam2 On to true
cond8	Siren On is true	Set smart door lock Lock to Unlock
cond9	Siren On is false	Set cam1 On to false Set cam2 On to false
cond10	Motion Detector 1 On is false	Set cam1 On to false
cond11	Motion Detector 2 On is false	Set cam2 On to false
cond12	Motion Detector 1 On is false	Set siren On to false
cond13	Motion Detector 2 On is false	Set siren On to false
cond14	Smart Door Lock Open is true	Set cam1 On to true
cond15	Window On is true	Set cam2 On to true

Figure 8 shows the complete set of 15 automation rules as configured in the Home Gateway's IoT automation interface.



Actions		Enabled	Name	Condition	Actions
Edit	Remove	Yes	cond1	motion detector1 On is true	Set siren On to true
Edit	Remove	Yes	cond2	motion detector2 On is true	Set siren On to true
Edit	Remove	Yes	cond3	smart door lock Open is true	Set siren On to true
Edit	Remove	Yes	cond4	window On is true	Set siren On to true
Edit	Remove	Yes	cond5	smoke detector Alarm is true	Set siren On to true
Edit	Remove	Yes	cond6	motion detector1 On is true	Set cam1 On to true
Edit	Remove	Yes	cond7	motion detector2 On is true	Set cam2 On to true
Edit	Remove	Yes	cond8	siren On is true	Set smart door lock Lock to Unlock
Edit	Remove	Yes	cond9	siren On is false	Set cam1 On to false Set cam2 On to false
Edit	Remove	Yes	cond10	motion detector1 On is false	Set cam1 On to false
Edit	Remove	Yes	cond11	motion detector2 On is false	Set cam2 On to false
Edit	Remove	Yes	cond12	motion detector1 On is false	Set siren On to false
Edit	Remove	Yes	cond13	motion detector2 On is false	Set siren On to false
Edit	Remove	Yes	cond14	smart door lock Open is true	Set cam1 On to true
Edit	Remove	Yes	cond15	window On is true	Set cam2 On to true

Figure 8: Complete Automation Rules Configured for SHIDS

4.3 Rule Details

4.3.1 Intrusion Alert Rules (cond1 – cond5)

Rules cond1 through cond5 ensure that any form of unauthorized access or hazardous condition immediately triggers the siren alarm. Motion detection at either zone (cond1, cond2), an opened smart door lock (cond3), an opened window (cond4), or a smoke/fire detection event (cond5) all independently activate the siren, providing comprehensive coverage across all monitored threat vectors.

4.3.2 Camera Activation Rules (cond6, cond7, cond14, cond15)

These rules ensure visual evidence is captured the moment any intrusion trigger occurs. Motion Detector 1 activates cam1 (cond6), and Motion Detector 2 activates cam2 (cond7). Additionally, an opened smart door lock activates cam1 (cond14), and an opened window activates cam2 (cond15), ensuring camera coverage is not limited to motion detection alone but extends to all physical entry point breaches.

Figure 9 illustrates the motion detection rule configuration that activates the siren and cameras upon motion trigger.

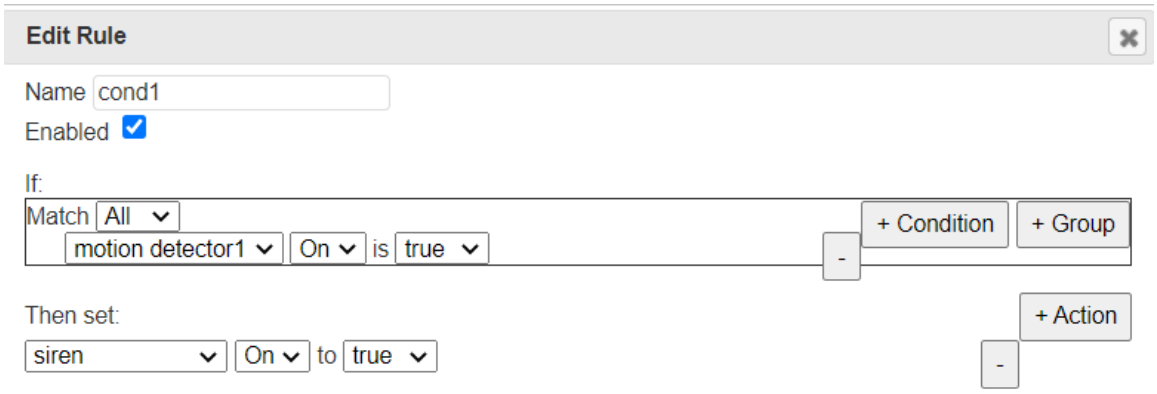


Figure 9: Motion Detection Rule for Alarm Activation

4.3.3 Access Control Rule (cond8)

Rule cond8 represents the system's automated containment response. When the siren is activated — regardless of which sensor triggered it — the smart door lock is automatically commanded to unlock. This rule models the emergency unlock scenario, allowing occupants to evacuate safely while simultaneously alerting the homeowner to the confirmed intrusion event.

Figure 10 illustrates the door lock control rule that automatically manages the smart lock state upon siren activation.

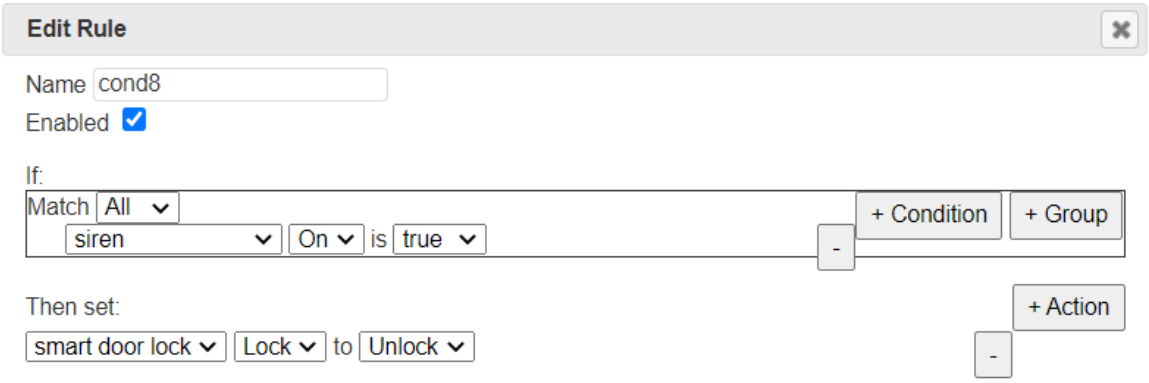


Figure 10: Automated Door Lock Control During Alarm Activation

4.3.4 Reset / Deactivation Rules (cond9 – cond13)

Rules cond9 through cond13 handle the system's return to its normal idle state once a threat condition has cleared. When the siren turns off (cond9), both cameras are deactivated simultaneously. When Motion Detector 1 returns to idle (cond10, cond12), cam1 and the siren are turned off respectively. Similarly, when Motion Detector 2 returns to idle (cond11, cond13), cam2 and the siren are deactivated. These reset rules ensure the system does not remain in a permanently triggered state and is always ready to detect the next intrusion event.

5 Testing and Results

This section presents the functional testing performed on the Smart Home Intrusion Detection System (SHIDS) simulation in Cisco Packet Tracer. The objective of testing was to verify that all configured automation rules respond correctly under simulated intrusion scenarios and that the system behaves as intended according to the design proposed in Part 1.

Several test cases were executed by manually triggering sensors and observing the corresponding automated system responses, including siren activation, camera activation, and smart device behaviour.

5.1 Testing Methodology

Testing was conducted by manually simulating intrusion or hazard events using the configured IoT sensors within Cisco Packet Tracer. For each test case:

- A sensor or device state was manually changed to trigger a specific event.
- The Home Gateway executed the corresponding automation rule(s).
- The output state of actuators and cameras was observed through the IoT Monitor interface and CCTV workstation.
- Results were recorded and validated against the expected behaviour.

The following scenarios were tested:

- Motion detected in Zone 1
- Motion detected in Zone 2
- Smart door lock opened / unlocked
- Window opened

5.2 Test Case 1: Motion Detector 1 Trigger

In this test case, **Motion Detector 1** was manually triggered to simulate movement at the primary monitored zone.

Expected behaviour:

- Siren activates
- Camera 1 activates

Observed result:

- Siren successfully activated
- Camera 1 successfully activated

Status: PASS

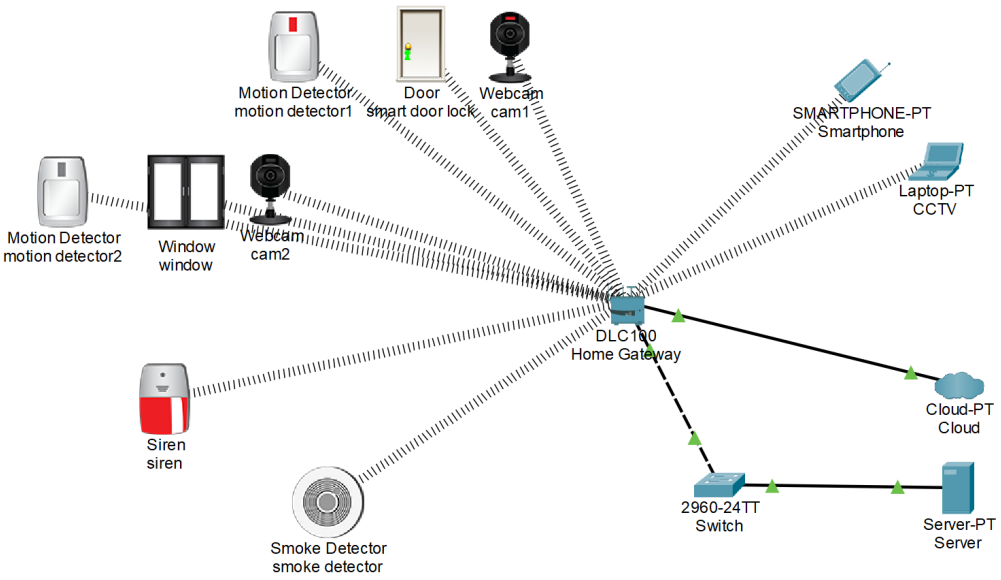


Figure 11: Motion Detector 1 Triggered: Siren and Camera 1 Activated

5.3 Test Case 2: Motion Detector 2 Trigger

In this scenario, **Motion Detector 2** was triggered to simulate motion detection in the secondary monitoring zone.

Expected behaviour:

- Siren activates
- Camera 2 activates

Observed result:

- Siren successfully activated
- Camera 2 successfully activated

Status: PASS

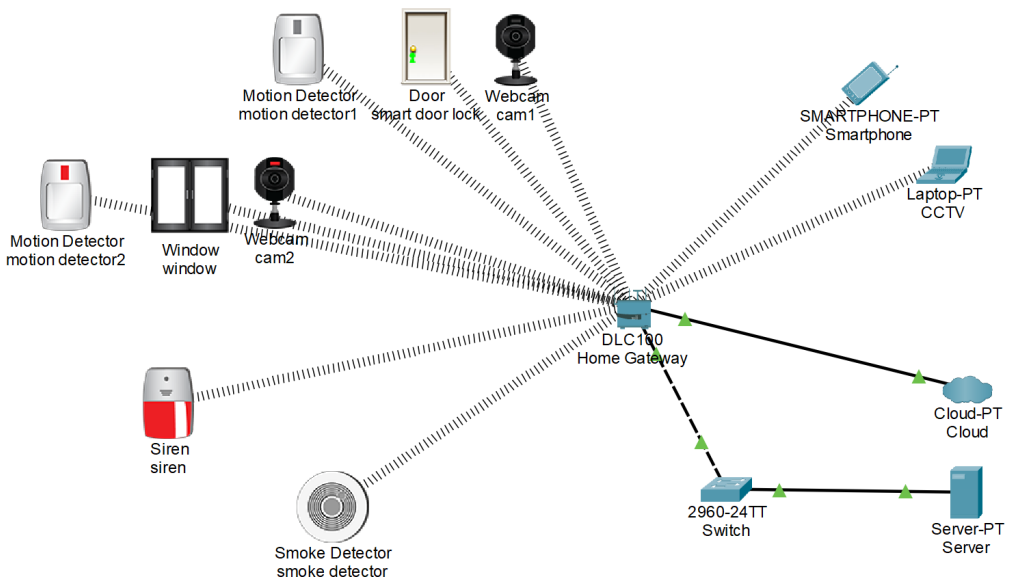


Figure 12: Motion Detector 2 Triggered: Siren and Camera 2 Activated

5.4 Test Case 3: Smart Door Unlock Trigger

This test simulates an unauthorized access attempt by manually unlocking the smart door lock.

Expected behaviour:

- Siren activates
- Camera 1 activates

Observed result:

- Siren successfully activated
- Camera 1 successfully activated

Status: PASS

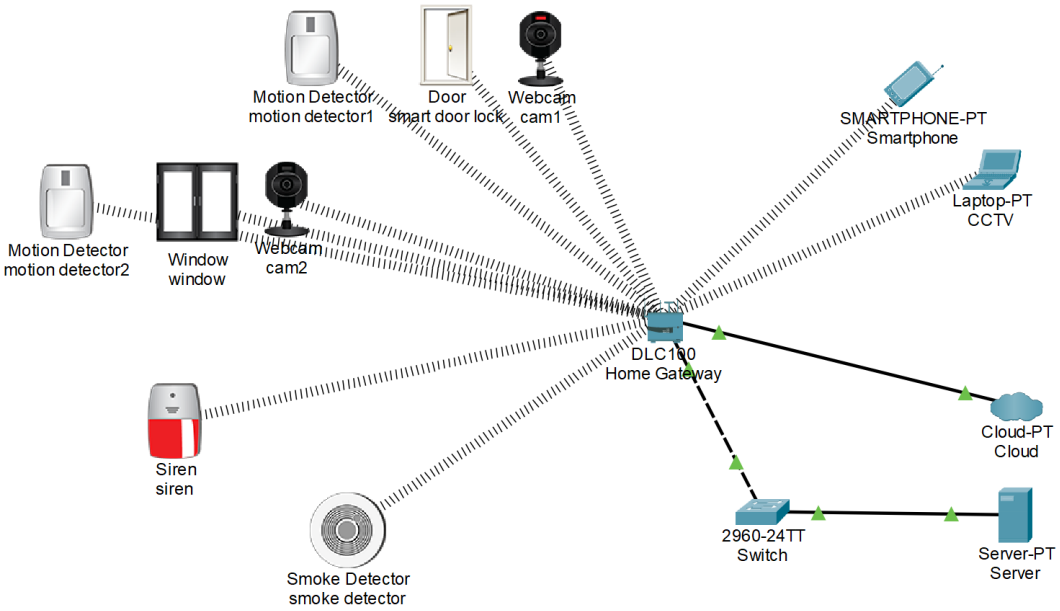


Figure 13: Smart Door Unlock Trigger: Siren and Camera 1 Activated

5.5 Test Case 4: Window Trigger

This scenario simulates an intrusion through the window sensor by manually opening the monitored window.

Expected behaviour:

- Siren activates
- Camera 2 activates

Observed result:

- Siren successfully activated
- Camera 2 successfully activated

Status: PASS

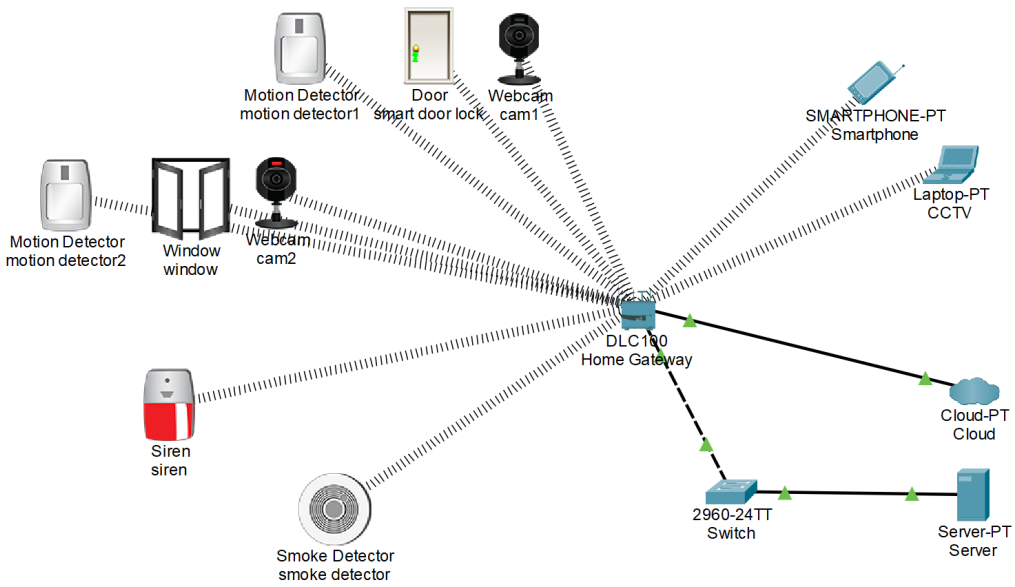


Figure 14: Window Trigger: Siren and Camera 2 Activated

5.6 Test Summary

Table 3 summarises all executed test cases and their results.

Table 3: SHIDS Functional Testing Summary

Test ID	Scenario	Expected Result	Status
TC1	Motion Detector 1 triggered	Siren + Camera 1 activated	PASS
TC2	Motion Detector 2 triggered	Siren + Camera 2 activated	PASS
TC3	Smart door unlocked	Siren + Camera 1 activated	PASS
TC4	Window opened	Siren + Camera 2 activated	PASS

All tested scenarios successfully produced the expected automated responses, confirming that the implemented automation rules and device configurations are functioning correctly. The SHIDS simulation therefore satisfies its primary security objectives, including intrusion detection, alert generation, and camera-based monitoring.

6 Challenges and Solutions

This section documents the primary technical challenges encountered during the implementation of the Smart Home Intrusion Detection System (SHIDS) prototype in Cisco Packet Tracer, along with the solutions applied to successfully complete the simulation.

Throughout the implementation process, several configuration and compatibility issues were encountered related to wireless connectivity, device registration, and network setup. Resolving these issues was necessary to ensure all devices could communicate correctly and participate in the automation workflow.

6.1 Laptop Wireless Connectivity Issue

One of the first challenges encountered was related to the CCTV monitoring laptop. By default, the selected laptop model in Cisco Packet Tracer did not include a wireless network interface, which prevented it from connecting to the Home Gateway over Wi-Fi.

Since the SHIDS design required the laptop to act as a wireless CCTV monitoring station, this issue prevented the device from accessing the IoT Monitor and viewing live camera feeds.

To resolve this issue, a **WPC300N wireless adapter** was manually installed through the laptop's Physical tab. After installation, the laptop gained access to a wireless interface and was successfully connected to the Home Gateway using the configured SSID and DHCP.

Figure 15 shows the wireless adapter installation performed on the laptop.

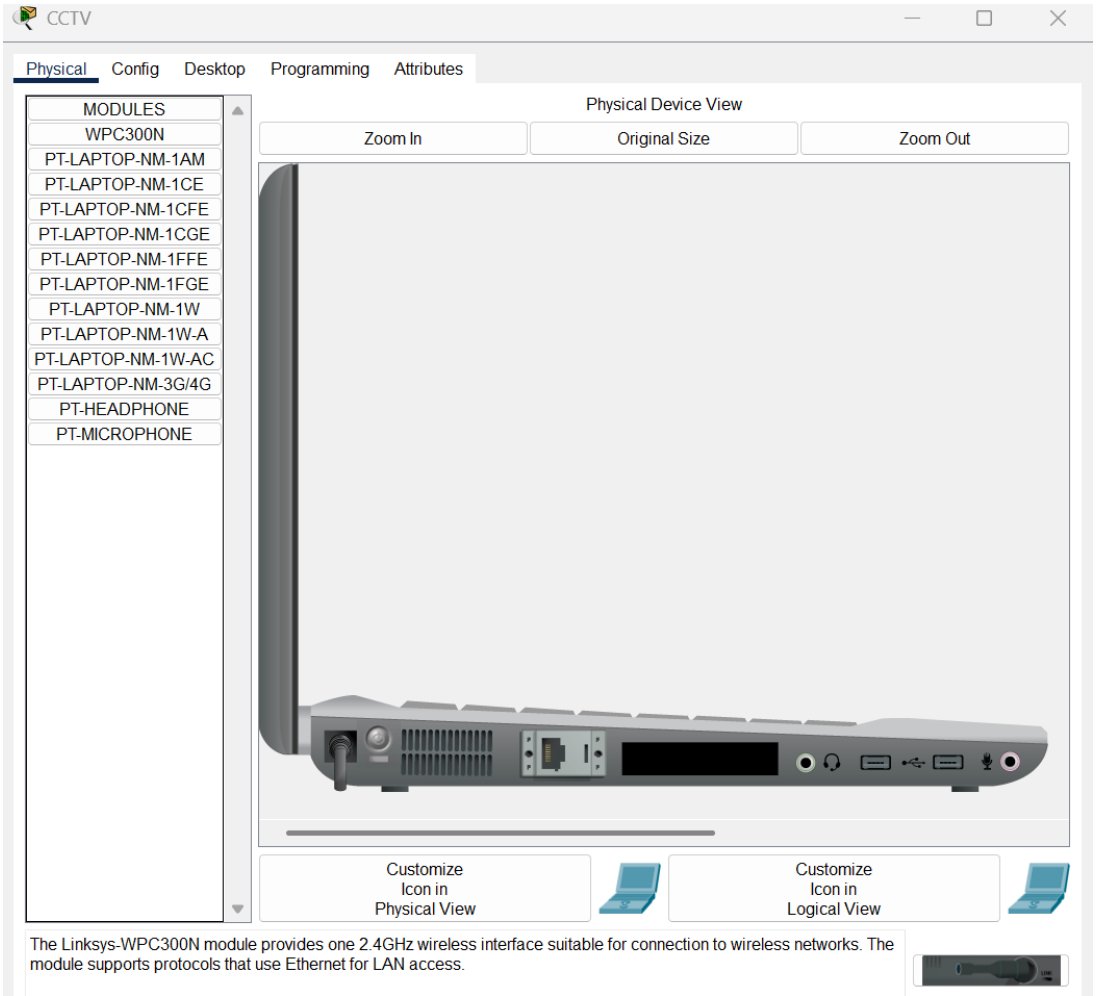


Figure 15: Installation of WPC300N Wireless Adapter on Laptop

6.2 IoT Device Registration Challenges

During initial setup, several IoT devices failed to appear in the IoT Monitor dashboard after being connected to the Home Gateway.

This issue was caused by incomplete device registration parameters, specifically missing or incorrect IoT server address configuration.

To resolve this issue:

- The Home Gateway's built-in IoT server was enabled.
- Each IoT device was manually configured to use the Home Gateway IP address as its IoT server.
- Default administrator credentials were used to authenticate and register devices successfully.

After reconfiguration, all devices appeared correctly in the IoT Monitor and became available for automation rule assignment.

6.3 Automation Rule Logic Verification

Another challenge involved validating the correctness of automation rules and ensuring no conflicting conditions existed between activation and deactivation logic.

For example, cameras and sirens needed to activate immediately when sensors were triggered, while also turning off correctly once sensors returned to idle.

Several rule iterations were tested and adjusted to ensure:

- Motion detectors correctly activate the corresponding camera.
- Intrusion events trigger the siren reliably.
- Cameras deactivate properly after the siren or motion event resets.
- Door and window triggers correctly activate alarm and surveillance responses.

After refinement, all automation rules passed functional testing successfully.

6.4 Summary of Challenges and Resolutions

Table 4 summarises the main implementation challenges and the solutions applied.

Table 4: Implementation Challenges and Solutions

Challenge	Solution Applied
Laptop lacked wireless connectivity	Installed WPC300N wireless adapter manually via Physical tab
IoT devices failed to register	Configured Home Gateway as IoT server and assigned correct server IP to devices
Automation rules initially inconsistent	Tested and refined trigger/deactivation logic until all conditions behaved correctly
Initial unfamiliarity with Packet Tracer IoT setup	Iterative testing and manual device configuration were performed until successful deployment

Overall, these challenges provided valuable practical experience in IoT device configuration, wireless networking, automation logic design, and simulation-based system troubleshooting.

7 Conclusion

This project successfully implemented a functional prototype of the **Smart Home Intrusion Detection System (SHIDS)** using Cisco Packet Tracer as an IoT simulation environment.

The simulation successfully integrated multiple IoT sensors, actuators, monitoring devices, and networking components into a unified smart home security architecture. All system components were configured and connected correctly through the Home Gateway, allowing centralized device management, wireless communication, and automation control.

The implemented automation rules were tested under multiple intrusion and hazard scenarios, including motion detection, door access events, window breaches, and smoke detection. In all test cases, the system responded as expected by activating the siren alarm, enabling the corresponding cameras, and executing automated access control actions.

The results confirm that the proposed SHIDS architecture is practical, functionally valid, and suitable for smart home security applications. Furthermore, this project demonstrates how IoT technologies can be leveraged to create intelligent, automated, and responsive residential security systems capable of real-time monitoring and threat response.

Overall, the successful completion of this prototype validates the design proposed in Part 1 and provides a strong foundation for future physical implementation or further expansion of the system with additional smart security features.